

# EN401 — Handout 3 Results & Analysis Report

## Base Power System Implementation & Backstop Penalty Verification

### 1. Solver Optimization Metrics

- **Status:** OPTIMAL | **Objective Value (NPV):** \$51,009,521.16 (~\$51.01M)
- **Problem Scale:** 466 Rows, 240 Columns, 2,160 Non-Zeros | **Runtime:** < 0.05s

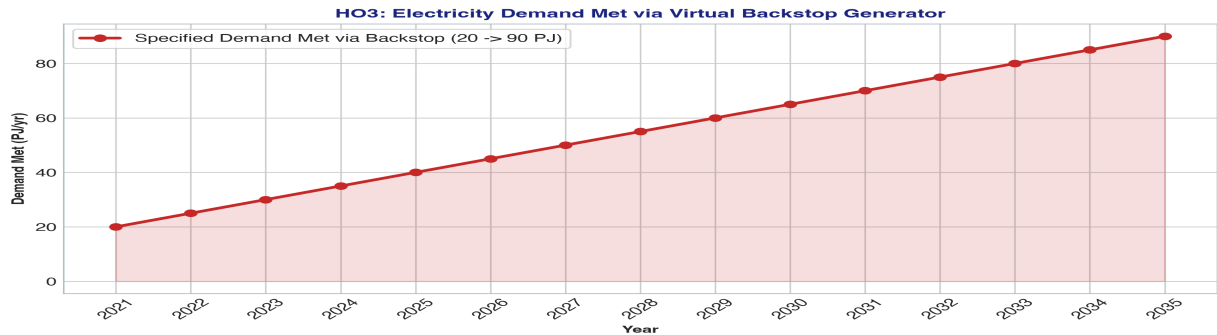


Figure: Electricity demand met via backstop generator.

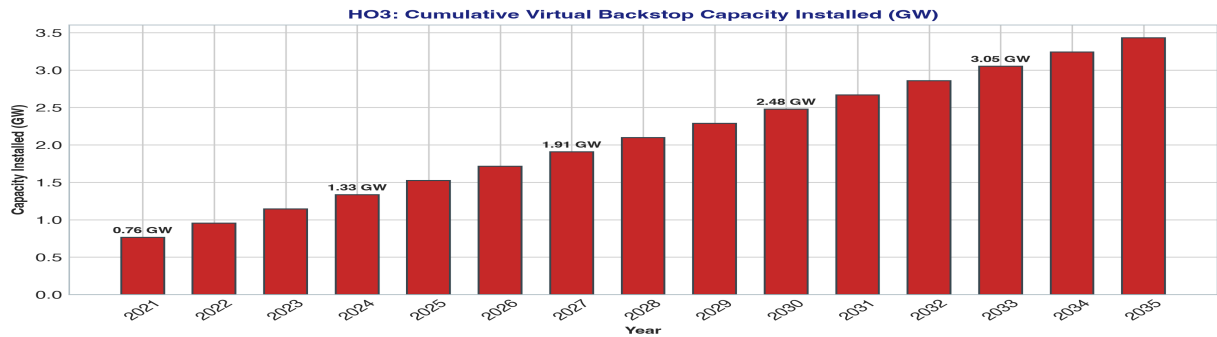


Figure: Backstop cumulative capacity expanding to 3.43 GW.

### 2. Numerical Optimization Results Table

| Metric            | 2021  | 2023  | 2025  | 2027  | 2029  | 2031  | 2033  | 2035  | 15-Yr Total |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------------|
| Demand Met (PJ)   | 20.0  | 30.0  | 40.0  | 50.0  | 60.0  | 70.0  | 80.0  | 90.0  | 825.0 PJ    |
| New BACKSTOP (GW) | 0.762 | 0.191 | 0.191 | 0.191 | 0.191 | 0.191 | 0.191 | 0.191 | 3.431 GW    |
| Cum BACKSTOP (GW) | 0.762 | 1.144 | 1.525 | 1.906 | 2.287 | 2.668 | 3.050 | 3.431 | 3.431 GW    |

### 3. Economic Interpretation

The extreme \$51.01M objective cost reflects an artificial penalty (\$99,999/kW) verifying supply adequacy and penalizing unmet demand.